# SCOPE ALPHA

## Technical Overview & Product Roadmap

### Trading Strategy Marketplace Platform

Demo Stage | March 2026

A crypto derivatives analytics and trading strategy marketplace that enables verified analysts to publish **timestamped trading signals**, create visual **chart annotations**, and monetize their expertise through **subscription tiers**.

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# 1. Executive Summary

Scope Alpha is a **crypto derivatives analytics and trading strategy marketplace**. The platform enables verified analysts to publish timestamped trading signals, create visual chart annotations, and monetize their trading expertise through tiered subscription plans.

Subscribers can browse the marketplace, view analyst profiles with **verified performance records**, and subscribe to receive real-time signals and chart annotations overlaid directly on their trading charts.

The application is currently in **demo/alpha stage** with mock data for development and public testing. The core platform architecture (authentication, signals, annotations, subscriptions, reviews, chat) has been built and is ready for iteration toward production.

| Demo Metrics | Value | Status |
|---|---|---|
| Total API Endpoints | 25+ | Implemented |
| Database Models | 10 | Production-ready |
| React Components | 30+ | Built |
| Pages / Routes | 15+ | Functional |
| Auth Providers | 3 (Email, Google, Demo) | Active |

# 2. Core Value Proposition

## For Analysts — Monetize Trading Expertise

- Publish timestamped, immutable trading signals that build a verified track record
- Create visual chart annotations (trend lines, zones, notes) for subscribers
- Build a subscriber base and earn recurring revenue through tiered subscriptions ($50–$2,000/mo)
- Performance metrics are auto-calculated — no manual reporting needed

## For Subscribers — Access Verified Strategies

- Browse analysts filtered by win rate, P&L, trading style, price, and rating
- Access verified trading strategies instead of trusting unverified social media callouts
- Get real-time signals and chart annotations directly on trading charts
- Leave reviews and ratings to help the community identify top performers

## For the Market — Accountability & Transparency

- All signals are timestamped and immutable after publication — no retroactive editing
- Performance metrics are auto-calculated from resolved trades, not self-reported
- Brings accountability to an industry plagued by unverifiable claims

# 3. Current Demo Features

The following features have been fully implemented in the demo. All features use mock/seeded data but are backed by **production-ready API routes and database models**.

## 3.1 Derivatives Terminal

The main trading dashboard provides a comprehensive view of crypto derivatives markets:

- **Price Charts** — TradingView-like charting powered by Lightweight Charts library
- **Health Score Dashboard** — Composite metric combining funding rates, open interest, liquidations, and basis
- **Funding Rate Analysis** — Current rates, 8-hour average, annualized carry, z-score, by-exchange breakdown
- **Open Interest Tracker** — Aggregate OI, net changes, per-exchange breakdown
- **Liquidation Monitor** — Long/short liquidation volumes with exchange heatmap
- **Perpetual Basis Analyzer** — Basis percentage, annualized rate, per-exchange comparison
- **Multi-Asset Screener** — Sortable table with sparklines across all metrics and regime classification

## 3.2 Analyst Marketplace

- Browse and search analysts by name
- Filter by subscription price, win rate, primary asset, subscriber count, and rating
- Sort by subscribers, win rate, P&L, price, rating, or newest
- Analyst cards display key metrics with performance sparklines
- Individual profile pages show full performance breakdown, active signals, and reviews

## 3.3 Signal System (Trading Callouts)

- Create signals with **asset, direction (long/short), entry/target/stop prices**
- Timeframe support: 1m, 5m, 15m, 1H, 4H, 1D, 1W
- Conviction levels: low, medium, high
- **Immutable publish timestamps** — signals cannot be edited after publication
- Validation logic ensures price coherence (longs: target > entry > stop; shorts: inverse)
- Manual resolution records actual closing price and calculates result P&L %

## 3.4 Chart Annotation System

- Horizontal trend lines with customizable labels
- Support/resistance zones rendered as colored areas
- Text notes placed at specific price levels
- Per-asset, per-timeframe storage with active/draft status

- Real-time sync across the UI via Zustand state management

## 3.5 Performance Analytics

All metrics are auto-calculated from resolved signals:

- Win rate, average P&L per trade, profit factor, risk/reward ratio
- Best and worst winning/losing streaks
- Total P&L and rolling 90-day P&L tracking
- Monthly returns chart and cumulative P&L curve visualization

## 3.6 Subscription & Monetization

- Analysts create 1–3 subscription tiers at different price points
- Each tier includes a description and feature list
- One active subscription per subscriber-analyst pair (uniqueness constraint)
- Stripe integration placeholder (mocked during demo, ready for production wiring)

## 3.7 Community Features

- **Chat** — Room-based messaging with reply threads and message pinning
- **Announcements** — Analyst broadcasts delivered to all subscribers
- **Reviews** — Subscriber ratings (1–5 stars) with text reviews and analyst responses

## 3.8 Authentication & Security

- **Email/Password** — Credentials provider with bcrypt password hashing
- **Google OAuth** — Optional social login via Google
- **Demo Mode** — One-click exploration without registration (auto-seeds analyst data)
- JWT session strategy with 30-day expiration
- Role-based access control: subscriber vs. analyst permissions on all endpoints

# 4. Tech Stack

| Layer | Technology |
| --- | --- |
| Frontend Framework | Next.js 14.2.5, React 18.3.1, TypeScript 5.4.5 |
| State Management | Zustand 4.5.2 with localStorage persistence |
| Styling | Tailwind CSS 3.4.4 with custom glass-morphism design system |
| Charts | Lightweight Charts 4.2.0 (TradingView-like) |
| Icons | Lucide React |
| Authentication | NextAuth 4.24.13, JWT sessions, bcrypt, Google OAuth |
| Database | PostgreSQL via Neon (cloud), connection pooling |
| ORM | Prisma 6.19.2 with full schema migrations |
| Fonts | Outfit (display), JetBrains Mono (code/data) |
| Build Tools | PostCSS 8.4.38, Autoprefixer 10.4.19 |
| Deployment | Vercel (frontend) + Neon PostgreSQL (database) |

# 5. Database Architecture

The database is modeled with Prisma ORM and deployed on Neon PostgreSQL. The schema consists of 10 interconnected models:

| Model | Description |
|---|---|
| User | Core account: email, password hash, display name, slug, role (subscriber/analyst), avatar, bio, social links |
| AnalystProfile | Extended profile: tagline, primary assets, trading style, experience years, subscription acceptance flag |
| AnalystPerformance | Calculated metrics: total signals, win count/rate, P&L; %, profit factor, Sharpe-like ratios, subscriber count |
| Signal | Trading callouts: asset, direction, entry/target/stop, timeframe, conviction, publish timestamp, result P&L; % |
| Annotation | Chart drawings: horizontal lines, zones, text notes tied to asset/timeframe with active/draft status |
| SubscriptionTier | Pricing tiers per analyst ($50-$2,000/mo) with feature lists |
| Subscription | Links subscribers to analysts: tier, period dates, Stripe integration fields |
| ChatMessage | Room-based messaging with reply threads and pinning support |
| Announcement | Analyst broadcasts delivered to subscribers |
| Review | Subscriber ratings (1-5 stars) with text content and analyst responses |

## Key Relationships & Constraints

- User 1:1 AnalystProfile (created on role upgrade)
- User 1:1 AnalystPerformance (auto-calculated from signals)
- AnalystProfile 1:N Signal, Annotation, SubscriptionTier, Announcement
- User (subscriber) N:M AnalystProfile via Subscription (unique per pair)
- User (subscriber) 1:N Review per analyst (one review per analyst)
- ChatMessage self-referencing for reply threads

# 6. API Architecture

All API routes follow RESTful conventions with role-based authorization. Mutation endpoints require authentication and verify user roles before processing.

| Group | Endpoint Pattern | Description |
|---|---|---|
| Auth | /api/auth/* | Signup, demo login, NextAuth (credentials + OAuth) |
| Users | /api/users/* | Profile management, become-analyst upgrade |
| Signals | /api/signals/* | CRUD with price validation, analyst-only mutations |
| Annotations | /api/annotations/* | CRUD for chart drawings, analyst-only mutations |
| Subscriptions | /api/subscriptions/* | Subscribe, manage, cancel (mock Stripe) |
| Tiers | /api/subscription-tiers/* | Create/fetch pricing tiers, analyst-only creation |
| Reviews | /api/reviews/* | Create reviews (subscriber-only), analyst responses |
| Chat | /api/chat/[roomId] | Room-based messaging with cursor pagination |
| Announcements | /api/announcements/* | Create/fetch broadcasts, analyst-only creation |
| Analysts | /api/analysts/[slug] | Public analyst profile lookup by slug |

## Authorization Pattern

Every mutation endpoint follows a consistent authorization check:

- Verify JWT session exists (401 Unauthorized if not)
- Check user role matches required permission (403 Forbidden if not)
- Validate request body against expected schema
- Execute database operation within Prisma transaction where needed

# 7. Product Roadmap

The following roadmap outlines the path from demo to production and beyond. Each phase builds on the previous, with Phase 1 being the minimum viable product.

## Phase 1 — Core Infrastructure (MVP)

Essential integrations needed to transition from demo to a functional product:

- **Real-time Market Data Feeds** — WebSocket integration with exchanges (Binance, Bybit, OKX) for live OHLC, funding rates, OI, and liquidation data
- **Stripe Payment Integration** — Subscription billing, analyst payouts, platform commission handling
- **Email Verification & Notifications** — Account verification, signal alerts, subscription confirmations
- **Real-time Signal Notifications** — Push notifications, email alerts, and Telegram bot integration for instant signal delivery

## Phase 2 — Enhanced Features

Feature enhancements that increase platform stickiness and value:

- **Copy Trading / Auto-Execution** — Allow subscribers to auto-mirror analyst trades on connected exchange accounts
- **Advanced Charting Tools** — Fibonacci retracements, trend channels, pattern recognition overlays
- **Portfolio Tracking** — Subscribers track their P&L from followed signals in a unified dashboard
- **Analyst Leaderboards** — Time-weighted rankings with multiple scoring criteria
- **Social Features** — Follow, like, share signals; social feed of analyst activity

## Phase 3 — Platform Growth

Expansion features for broader market reach:

- **Mobile App** — React Native cross-platform app for iOS and Android
- **API Access** — Programmatic signal consumption for algorithmic traders and third-party integrations
- **Custom Indicator Marketplace** — Pine Script equivalent for creating and selling custom indicators
- **Research Publishing** — Platform for analysts to publish long-form research papers and market analysis
- **Backtesting Engine** — Historical strategy validation against real market data
- **Multi-Asset Expansion** — Extend beyond crypto to forex, stocks, commodities, and indices

## Phase 4 — Enterprise

Institutional-grade features for professional firms:

- **Institutional Verification** — KYC/AML compliance for analyst identity verification
- **White-Label Solutions** — Customizable platform for trading firms and hedge funds

- **Audit Trail & Compliance** — Full reporting for regulatory requirements
- **Admin Analytics Dashboard** — Platform-wide metrics, user behavior, and revenue analytics

# 8. Revenue Model

| Revenue Stream | Description | Priority |
|---|---|---|
| Subscription Commission | 15–25% of all analyst subscription revenue | Primary |
| Premium Listings | Paid placement for analyst visibility in marketplace | Secondary |
| Featured Promotions | Analysts pay for featured placement and boosted visibility | Secondary |
| API Access Tiers | Paid tiers for programmatic signal access (developers, algo traders) | Future |
| Enterprise Licensing | White-label platform licensing for trading firms | Future |

The primary revenue driver is the platform commission on analyst subscriptions. With analyst tiers ranging from $50 to $2,000/month, even a modest analyst base generates meaningful recurring revenue. Secondary streams diversify income as the platform scales.

# 9. Competitive Advantages

- **Immutable, Timestamped Signals** — Unlike social media callouts, signals cannot be retroactively edited or deleted. This creates genuine accountability.
- **Transparent Performance Metrics** — All metrics are auto-calculated from resolved trades, eliminating self-reported and potentially fabricated performance claims.
- **Professional-Grade Charting** — Integrated charting with analyst annotation overlay, not just text-based signals. Subscribers see exactly what the analyst sees.
- **Derivatives-Focused** — Purpose-built for derivatives trading with funding rates, open interest, liquidations, and basis analysis. Not a generic signals platform.
- **Modern Tech Stack** — Next.js 14, React 18, TypeScript, Prisma, Zustand — enabling rapid iteration and a responsive user experience.
- **Glass-Morphism UI** — A distinctive, modern design language that differentiates from legacy trading platforms and signals services.

# 10. Current Limitations (Demo State)

The following limitations are expected for a demo-stage product and represent the gap between the current build and a production-ready platform:

| Area | Limitation | Resolved In |
|---|---|---|
| Market Data | Mock/generated data, not live exchange feeds | Phase 1 |
| WebSocket Feeds | No live real-time data streaming | Phase 1 |
| Payments | Stripe not connected; subscriptions are mocked | Phase 1 |
| Email System | No email verification or notifications | Phase 1 |
| State Persistence | Some features use local state (Zustand) instead of DB | Phase 1 |
| Notifications | No push, email, or Telegram signal alerts | Phase 1 |
| Mobile | Basic responsive only; no dedicated mobile app | Phase 3 |

# 11. Conclusion

Scope Alpha demonstrates a **functional proof-of-concept** for a trading strategy marketplace. The core platform architecture — authentication, trading signals, chart annotations, subscriptions, reviews, and community chat — has been built with production-quality code patterns.

The **database schema, API routes, and frontend components** are all designed for scalability and follow modern best practices (TypeScript, Prisma ORM, JWT auth, role-based access control). The glass-morphism design system provides a distinctive, professional visual identity.

The immediate next steps focus on **Phase 1** of the roadmap: integrating real-time market data feeds, connecting Stripe for payment processing, implementing email notifications, and building real-time signal delivery. These integrations will transition Scope Alpha from a demo into a functional product ready for early users and market validation.